Input-output table development

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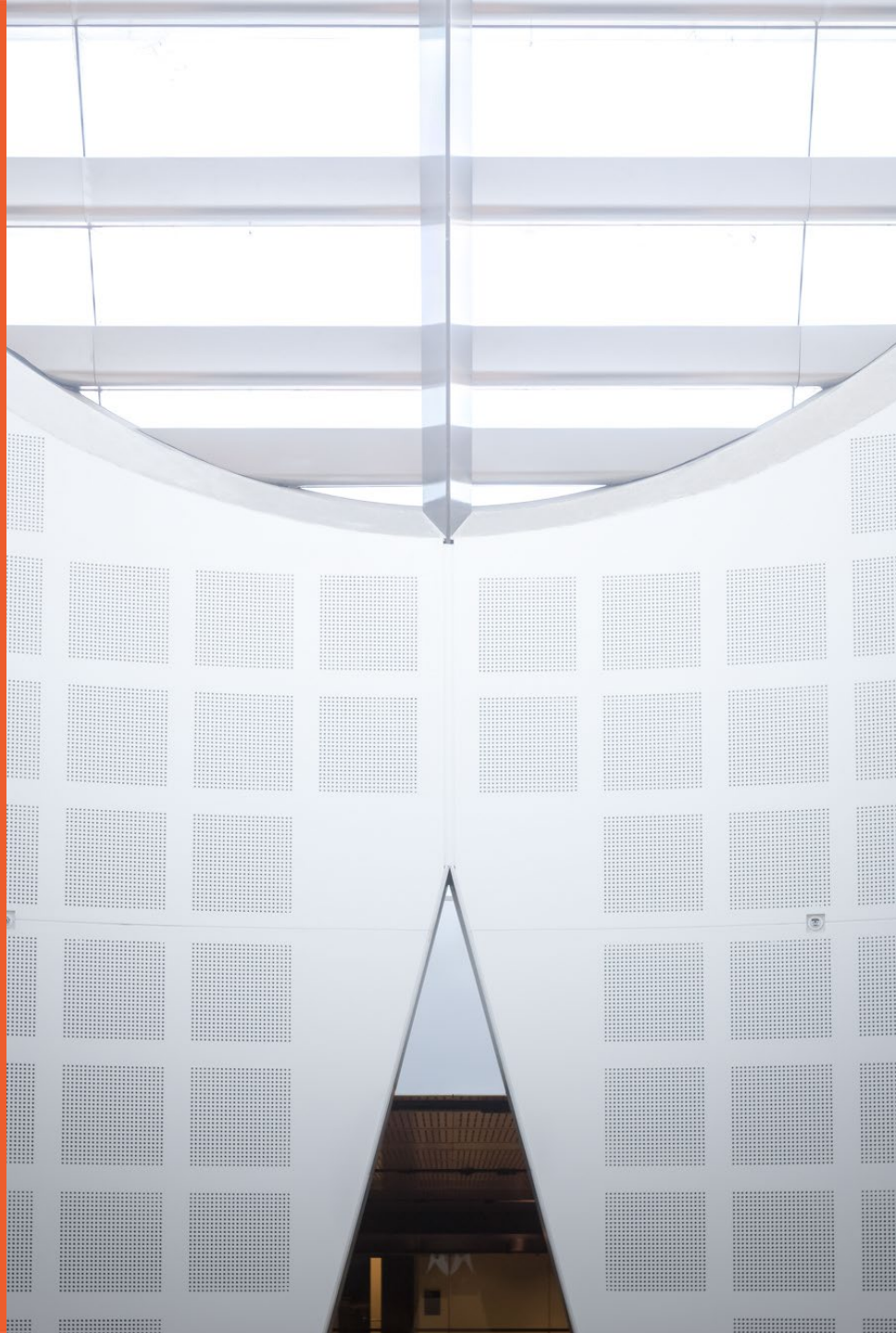
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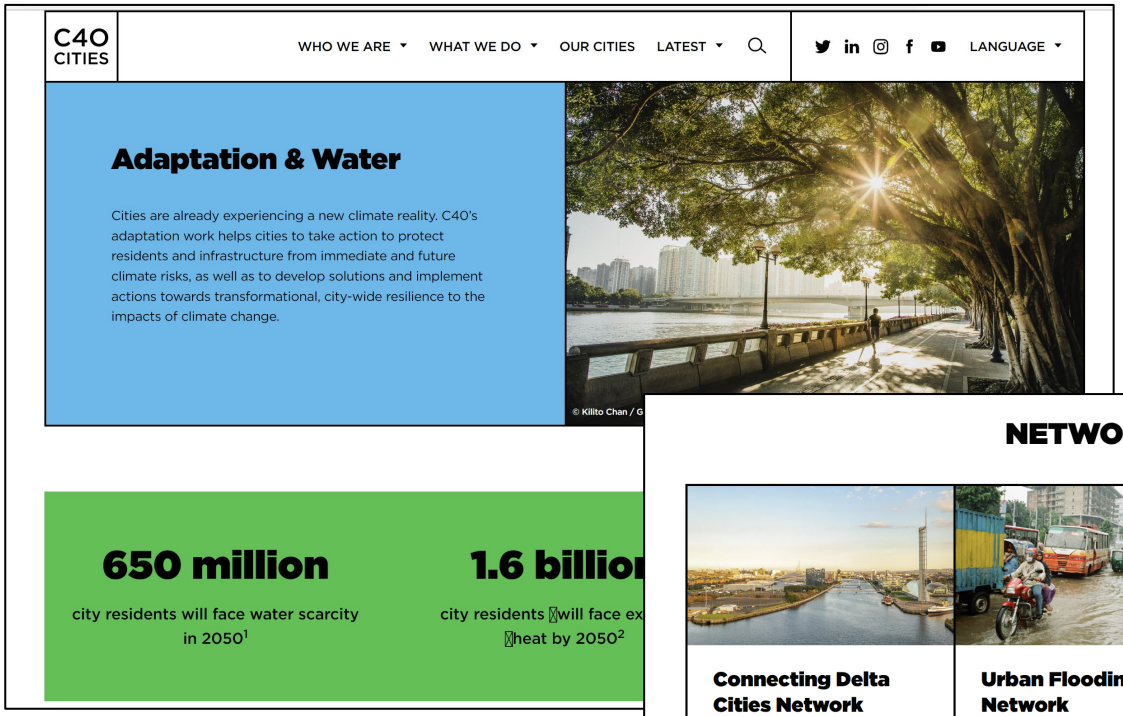
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Recap – Week 2

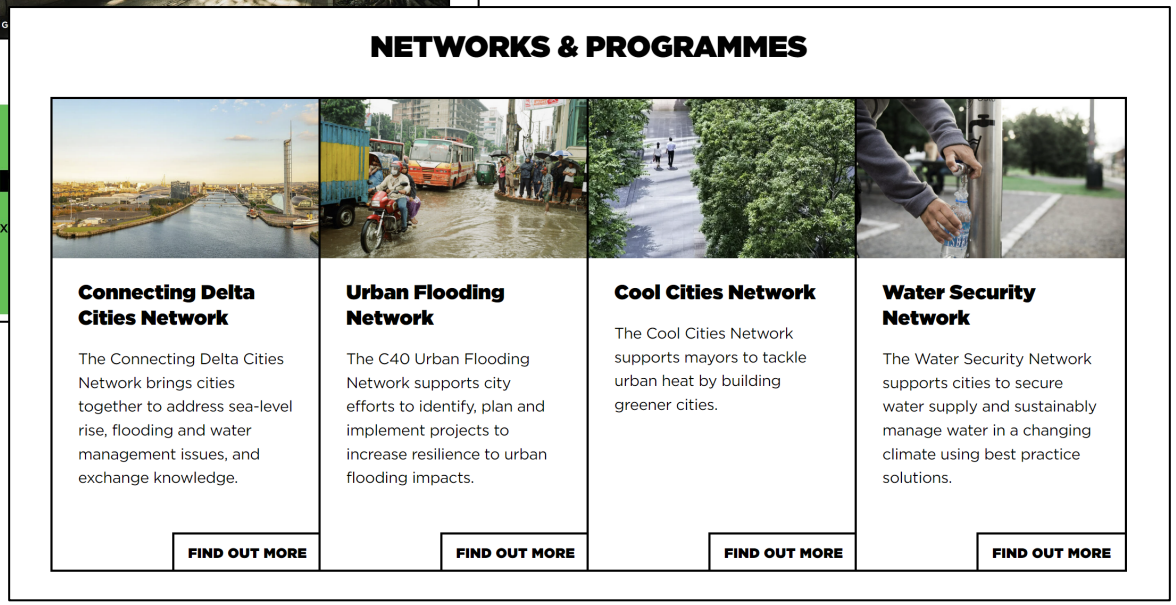
- Matrix notation
- Matrix properties
- Matrix multiplication
- Matrix summation operators
- Balance your own input-output table

Who is interested in environmental footprinting?

Local Government

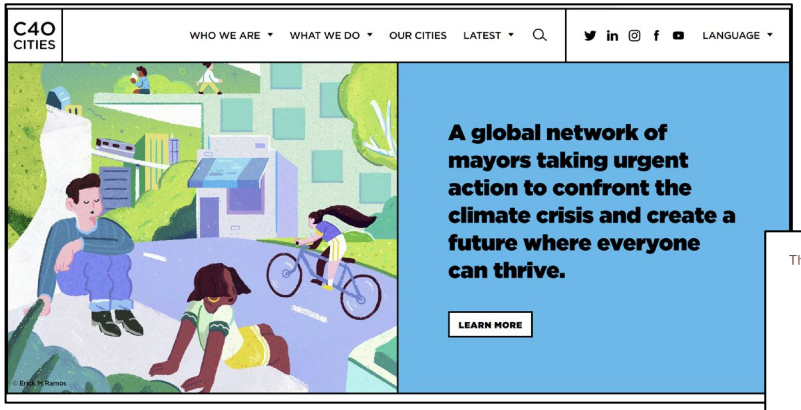


https://www.c40.org/what-we-do/scaling-up-climate-action/adaptation-water/

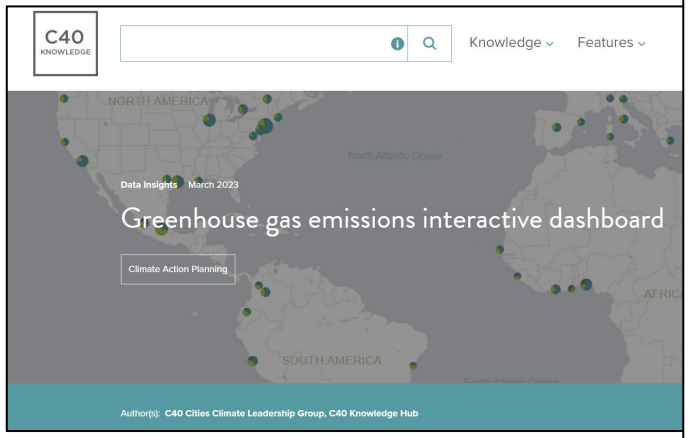


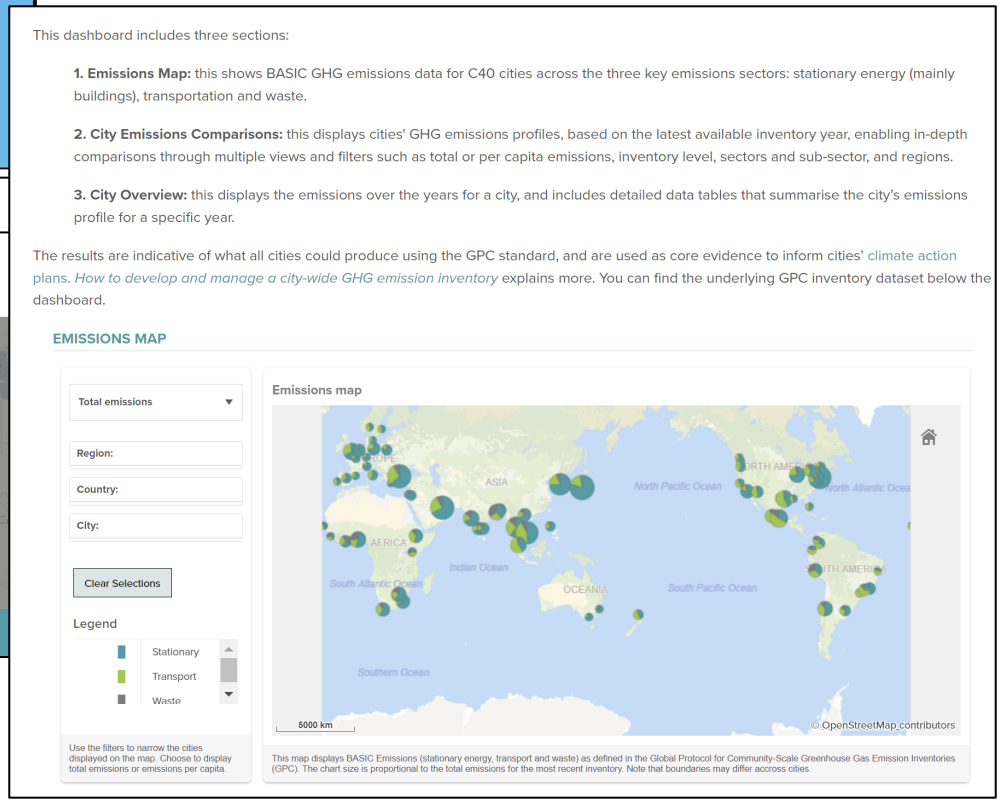
Who is interested in environmental footprinting?

Local Government

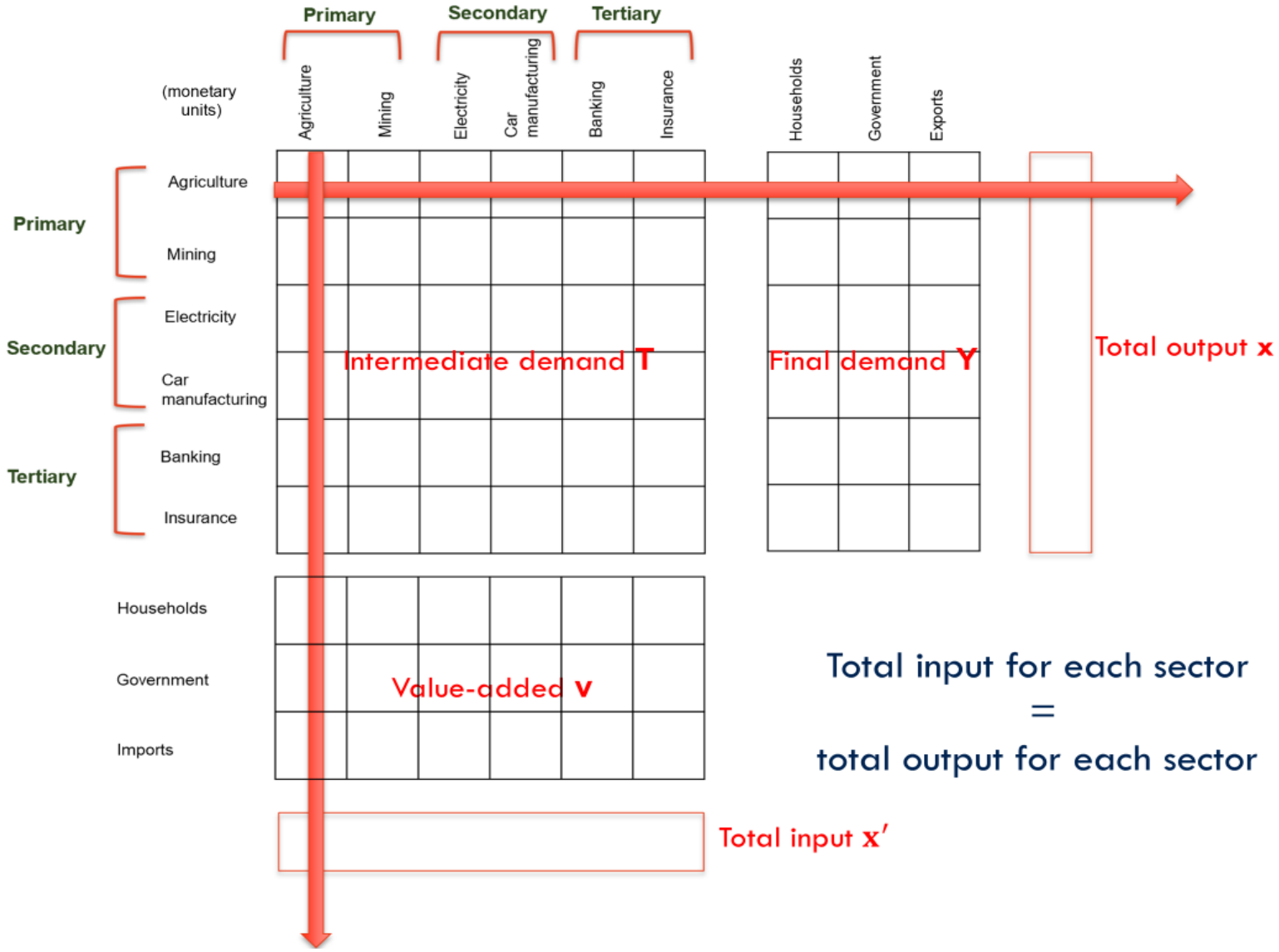


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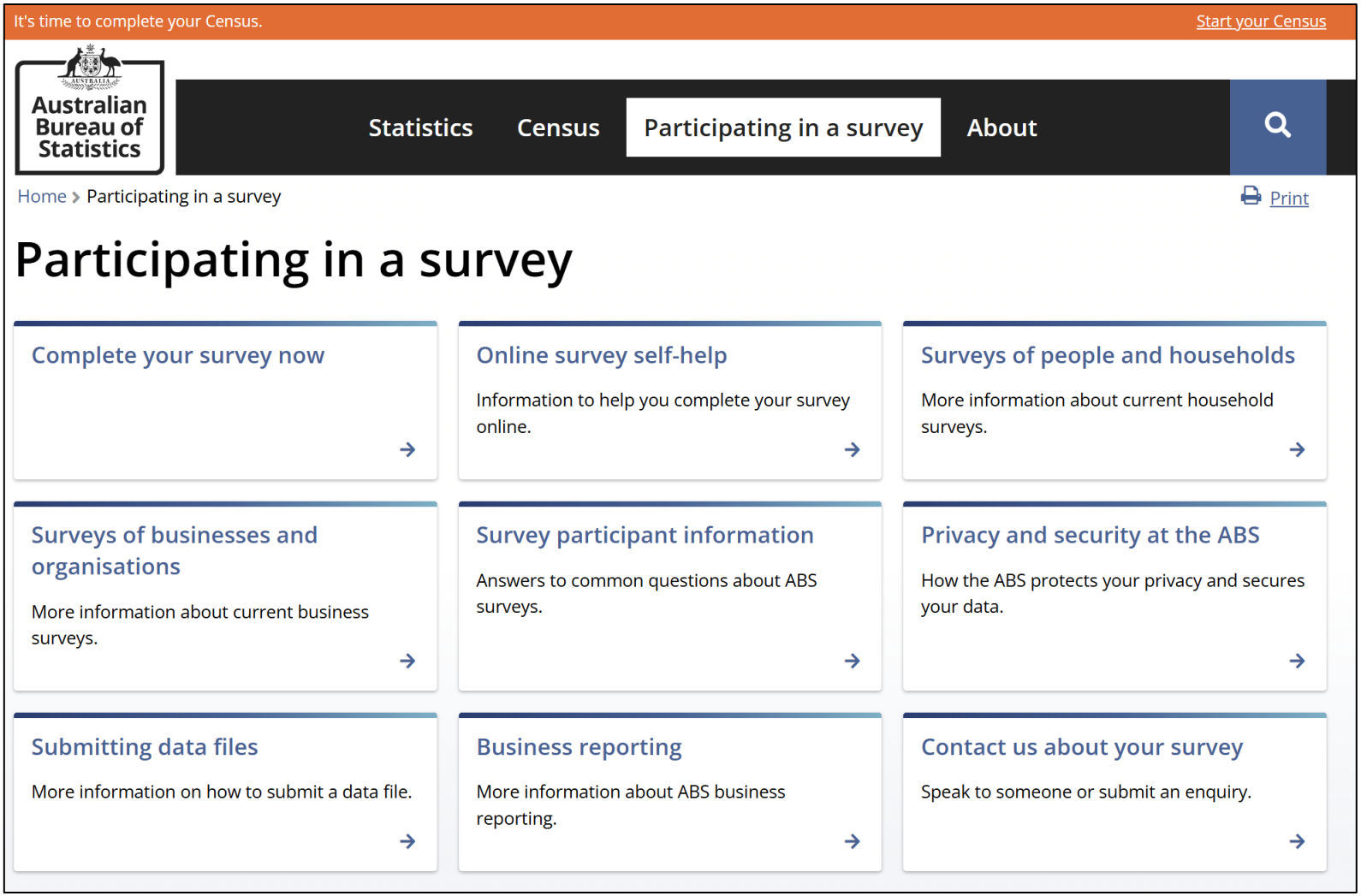




Input-output tables

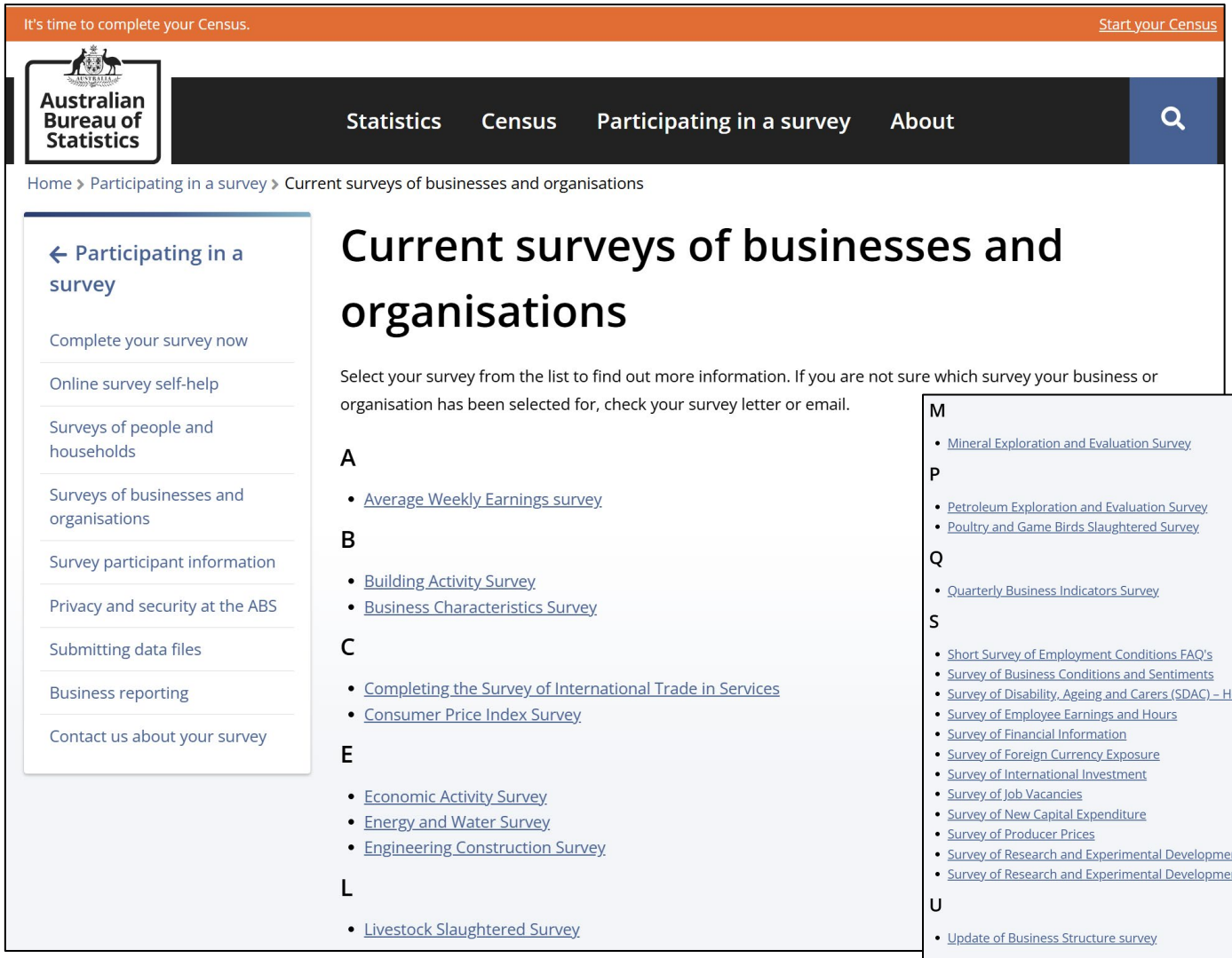


Where are the data sourced from?



https://www.abs.gov.au/participate-survey

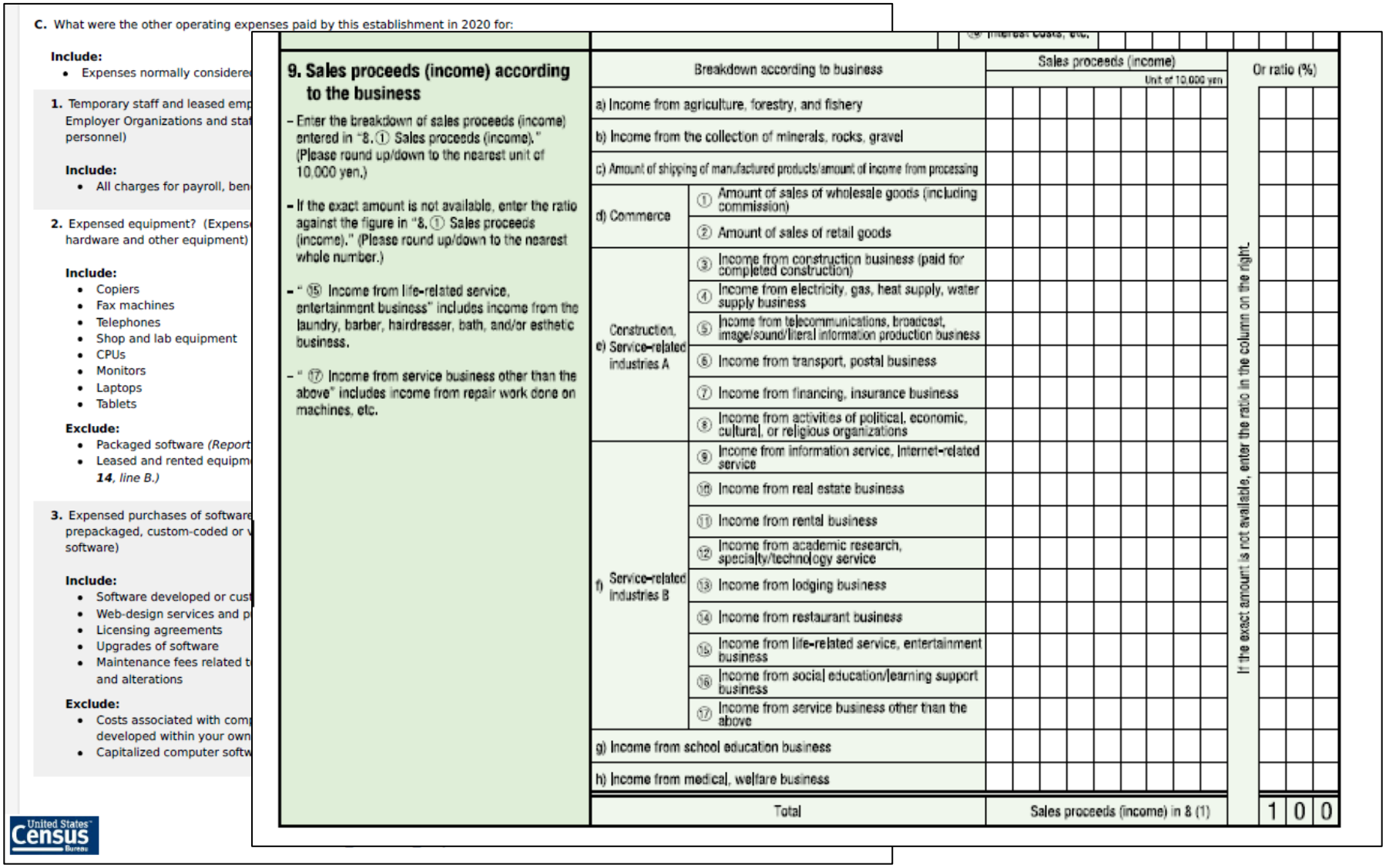
Where are the data sourced from?





https://www.abs.gov.au/participate-survey/business-survey

Where are the data sourced from?



C. What were the other operating expenses paid by this establishment in 2020 for:

Include:
- Expenses normally considere

1. Temporary staff and leased emp
Employer Organizations and stat
personnel)

Include:
- All charges for payroll, ben

2. Expensed equipment? (Expens
hardware and other equipment)

Include:
- Copiers
- Fax machines
- Telephones
- Shop and lab equipment
- CPUs
- Monitors
- Laptops
- Tablets

Exclude:
- Packaged software *(Report*
- Leased and rented equipm
***14**, line B.)*

3. Expensed purchases of software
prepackaged, custom-coded or
software)

Include:
- Software developed or cust
- Web-design services and p
- Licensing agreements
- Upgrades of software
- Maintenance fees related t
and alterations

Exclude:
- Costs associated with com
developed within your own
- Capitalized computer softw

9. Sales proceeds (income) according to the business

– Enter the breakdown of sales proceeds (income) entered in "8. ① Sales proceeds (income)." (Please round up/down to the nearest unit of 10,000 yen.)

– If the exact amount is not available, enter the ratio against the figure in "8. ① Sales proceeds (income)." (Please round up/down to the nearest whole number.)

– "⑮ Income from life-related service, entertainment business" includes income from the laundry, barber, hairdresser, bath, and/or esthetic business.

– "⑰ Income from service business other than the above" includes income from repair work done on machines, etc.

Breakdown according to business		Sales proceeds (income) Unit of 10,000 yen	Or ratio (%)
a) Income from agriculture, forestry, and fishery			
b) Income from the collection of minerals, rocks, gravel			
c) Amount of shipping of manufactured products/amount of income from processing			
d) Commerce	① Amount of sales of wholesale goods (including commission)		
	② Amount of sales of retail goods		
e) Construction, Service-related industries A	③ Income from construction business (paid for completed construction)		
	④ Income from electricity, gas, heat supply, water supply business		
	⑤ Income from telecommunications, broadcast, image/sound/literal information production business		
	⑥ Income from transport, postal business		
	⑦ Income from financing, insurance business		
	⑧ Income from activities of political, economic, cultural, or religious organizations		
f) Service-related industries B	⑨ Income from information service, Internet-related service		
	⑩ Income from real estate business		
	⑪ Income from rental business		
	⑫ Income from academic research, specialty/technology service		
	⑬ Income from lodging business		
	⑭ Income from restaurant business		
	⑮ Income from life-related service, entertainment business		
	⑯ Income from social education/learning support business		
	⑰ Income from service business other than the above		
g) Income from school education business			
h) Income from medical, welfare business			
Total		Sales proceeds (income) in 8 (1)	1 0 0

If the exact amount is not available, enter the ratio in the column on the right.

United States Census Bureau

The System of National Accounts (SNA)

"The System of National Accounts (SNA) is the internationally agreed standard set of recommendations on how to compile measures of economic activity in accordance with strict accounting conventions based on economic principles"

"They provide a comprehensive and detailed record … conomic activities taking place within an economy and of the i… fferent economic agents, and groups of agents, that tak… The framework of the SNA provides accounts th…

a) comprehen… … ror all agents in an e…
b) con… …ish the consequences of a single actic… … accounting rules;
c) integr… …nces of a single action by one agent are necess… …uiting accounts, including the impact on measurement of wealth … …ance sheets."

The accounting rules under the SNA are different from those used in business accounting. One set of rules is used to account for a country's economic activity, the other is used to account for an entity's economic activity.

For more detail see: https://unstats.un.org/unsd/nationalaccount/docs/SNA2008.pdf

Calculation of Gross Domestic Product

GDP represents the sum of all economic activity within a country

1. Production approach

GDP = total value of goods and services produced
- intermediate consumption

2. Income approach

GDP = compensation of employees
+ gross operating surplus
+ gross mixed income
+ taxes
- subsidies

3. Expenditure approach

GDP = final consumption
+ gross capital formation
+ exports of goods and services
+ changes in inventories
- imports of goods and services

Calculation of Gross Domestic Product

	A	B	DN	DO	DP	DQ	DR	DS	DT	DU	DV	DW
1		**Use**										
2		Australian Input-Output Table 2023-24 (Table 2) https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-input-output-tables/latest-release#data-downloads	Total Industry Uses	Households ; Final Consumption Expenditure	General Government ; Final Consumption Expenditure	Private ; Gross Fixed Capital Formation	Public Corporations ; Gross Fixed Capital Formation	General Government ; Gross Fixed Capital Formation	Changes in Inventories	Exports of Goods and Services	Final Uses (Q1 to Q7)	Total Supply
3	**Supply**		T4	Q1	Q2	Q3	Q4	Q5	Q6	Q7	T5	T6
4	0101	Sheep, grains, beef and dairy cattle	32 257	161	0	2 493	0	0	- 948	17 327	19 033	51 290
5	0102	Poultry and other livestock	8 897	1 246	0	1 549	0	0	34	293	3 122	12 019
6	0103	Other agriculture	22 928	12 709	0	724	0	0	- 489	2 680	15 624	38 552
7	0201	Aquaculture	2 086	1 196	0	0	0	0	76	72	1 344	3 430
8	0301	Forestry and logging	3 944	128	82	0	0	0	71	185	466	4 410
9	0401	Fishing, hunting and trapping	1 265	866	28	0	0	0	44	821	1 759	3 024
10	0501	Agriculture, forestry and fishing support services	11 792	0	869	0	0	0	- 90	4 077	4 856	16 648
11	0601	Coal mining	5 202	0	0	0	0	0	- 234	87 669	87 435	92 637
12	0701	Oil and gas extraction	20 045	6 277	0	0	0	0	28	81 436	87 740	107 785
13	0801	Iron ore mining	2 236	0	0	0	0	0	- 496	132 584	132 089	134 325
14	0802	Non-ferrous metal ore mining	51 038	0	0	0	0	0	603	18 427	19 030	70 068
15	0901	Non-metallic mineral mining	7 917	20	0	0	0	0	489	10 233	10 742	18 659
16	1001	Exploration and mining support services	25 897	0	274	5 481	0	0	0	240	5 995	31 892
17	1101	Meat and meat product manufacturing	19 498	12 448	0	0	0	0	250	17 497	30 195	49 693
18	1102	Processed seafood manufacturing	1 392	2 531	0	0	0	0	- 75	352	2 808	4 200
19	1103	Dairy product manufacturing	8 181	10 343	0	0	0	0	254	3 226	13 824	22 005
20	1104	Fruit and vegetable product manufacturing	1 911	5 741	0	0	0	0	17	3 173	8 931	10 842
21	1105	Oils and fats manufacturing	1 847	1 053	0	0	0	0	16	624	1 693	3 540

18	9502	Other services	2 034
19			
20		**Total Intermediate Use**	**2 433 987**
21	P1	Compensation of employees	1 269 749
22	P2	Gross operating surplus & mixed income	1 150 716
23	P2.1	Gross operating surplus	974 976
24	P2.2	Gross mixed income	175 740
25	P3	Taxes less subsidies on products	34 128
26	P4	Other taxes less subsidies on production	90 887
27	P5	Complementary imports	0
28			
29		**Australian Production**	**4 979 467**
30	P6	Competing imports	589 795
31			
32	T3	Total Uses	5 569 262
33			
34	V1	Gross value added (P1+P2+P4)	2 511 352
35	**GDP**	**Gross domestic product (Total V1+Total P3)**	
36			
37		© Commonwealth of Australia 2026	

Final demand (Y)

Value added (V)

Units: AUD million

https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-input-output-tables/latest-release#data-downloads

Calculation of Gross Domestic Product

	A	B	DV	DW
1		**Use**		
2		Australian Input-Output Table 2023-24 (Table 2) https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-input-output-tables/latest-release#data-downloads	Final Uses (Q1 to Q7)	Total Supply
3	**Supply**		T5	T6
118	9502	Other services	13 477	15 511
119				
120		**Total Intermediate Use**	**3 135 275**	**5 569 262**
121	P1	Compensation of employees	0	1 269 749
122	P2	Gross operating surplus & mixed income	0	1 150 716
123	P2.1	Gross operating surplus	0	974 976
124	P2.2	Gross mixed income	0	175 740
125	P3	Taxes less subsidies on products	134 545	168 673
126	P4	Other taxes less subsidies on production	0	90 887
127	P5	Complementary imports	0	0
128				
129		**Australian Production**	**3 269 820**	**8 249 287**
130	P6	Competing imports	13 850	603 645
131				
132	T3	Total Uses	3 283 670	8 852 932
133				
134	V1	Gross value added (P1+P2+P4)		
135	**GDP**	**Gross domestic product (Total V1+Total P3)**		**2 680 025**
136				
137		© Commonwealth of Australia 2026		

Income approach

$1,269,749 +
$1,150,716 +
$168,673 +
$90,887
$2,680,025

Production approach

$8,249,287 -
$5,569,262
$2,680,025

Expenditure approach

$3,283,670 -
$603,645
$2,680,025

Units: AUD million

SNA Challenges

What if the three approaches don't reconcile?

How do you capture transactions that are "off the books"?

Madagascar employment data

Table 5.6.3: Calculated employment matrix

Activities	Formal Sector		Administration	Household Sector		Total Employments
	Formal Employment	Informal Employment		Informal Employment	Non Informal Employment	
Primary	78 917	5 245	-	1 915 995	-	2 000 157
Extractive	14 898	-	-	56 666	-	71 564
Food	30 216	-	-	101 772	-	131 988
Confection	67 985	490	-	234 537	-	303 012
Wood	3 209	-	-	12 207	-	15 416
Other Industries	37 635	-	-	143 153	-	180 788
Energy	7 207	320	-	-	-	7 527
Construction	32 733	236	-	112 925	-	145 894
Trade	64 625	3 993	-	748 804	-	817 422
Transport	12 002	1 771	-	148 889	-	162 662
Household	7 360	-	-	-	234 618	241 978
Education	38 881	9 460	133 951	63 212	-	245 505
Health	17 058	704	16 373	27 762	-	61 897
Other Services	95 524	331	-	216 218	-	312 073
Public Administration	-	-	124 650	-	-	124 650
TOTAL	508 250	22 550	274 974	3 782 141	234 618	4 822 532

Source: United Nations Economic Commission for Africa. Operational Guidebook on Accounting for Informal Sector in National Accounts (**operational_guidebook.pdf (unescwa.org)**)

Informal economy

Box 1.1: Areas of possible under-estimation in GDP

(ICP 2011 Global Office, Operational Material: National Accounts Framework in the ICP)

In ensuring that the GDP estimates are exhaustive, national accountants should look closely at the following areas, which experience has shown can be under-estimated in (or even omitted from) countries' accounts:

- consumption of crops and livestock products that households have produced for themselves;
- food, clothing and household goods sold by street traders or in village markets;
- goods and services that are provided to households by "informal" or "unregistered" producers, such as:
  - food and drinks sold on the street by itinerant vendors;
  - services of unregistered taxi drivers;
  - plumbing, electrical, repairing vehicles, and other household maintenance services;
  - house-cleaning and child-minding;
  - private teaching lessons; and
  - personal services such as hair-cutting and shoe-cleaning;
- goods that have been smuggled into the country from abroad without payment of customs duties and without being recorded in the international trade statistics;
- all government expenditures, including expenditures on the military forces, expenditures by municipal and local authorities, and expenditures by the head of state;
- dwellings that people build for themselves; these may be constructed with traditional materials such as sun-baked rather than fired bricks and with palm fronds and similar thatching materials but they nevertheless constitute gross fixed capital formation and they also provide dwelling services to the owner-occupiers, which are included in household final consumption expenditure;
- illegal activities that contribute significantly to production, such as prostitution and drugs.

Source: United Nations Economic Commission for Africa. Operational Guidebook on Accounting for Informal Sector in National Accounts (**operational_guidebook.pdf (unescwa.org)**)

Interesting insights from input-output tables

What total output is required from the 'Basic chemical manufacturing' sector if the final demand for the 'Other food product manufacturing' sector doubles?

What impact will a drought have on the economy assuming a 20% decline in output from the 'Sheep, grains, beef and dairy cattle' sector?

	Sheep, grains, beef and dairy cattle	Other food product manufacturing	Basic chemical Manufacturing	All Others	Final Demand	Total output
Sheep, grains, beef and dairy cattle	2,585	143	62	28,012	10,664	41,466
Other food product manufacturing	765	1,060	20	5,892	9,981	17,718
Basic chemical Manufacturing	1,619	56	1,456	18,148	8,551	29,828
All Others	21,911	7,223	7,798	1,728,598	2,232,714	3,953,548
…	…	…	…	…	…	…

https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-input-output-tables/2018-19/5209055001DO001_201819.xls

The impact of a cyclone

Nat. Hazards Earth Syst. Sci., 19, 137–151, 2019
https://doi.org/10.5194/nhess-19-137-2019
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Natural Hazards and Earth System Sciences
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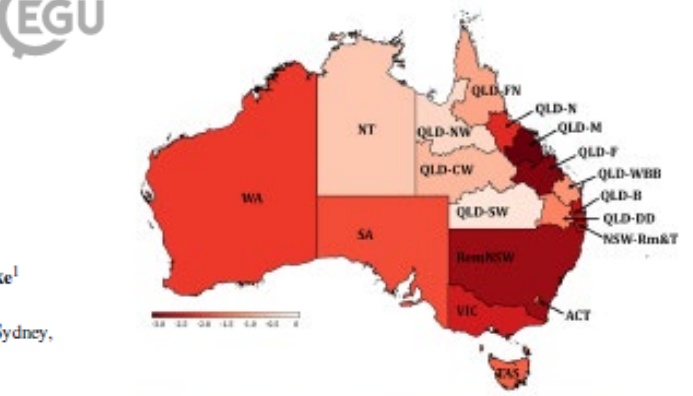
Economic damage and spillovers from a tropical cyclone

Manfred Lenzen[1], Arunima Malik[1,2], Steven Kenway[3], Peter Daniels[4], Ka Leung Lam[3], and Arne Geschke[1]

[1]ISA, School of Physics A28, The University of Sydney, NSW, 2006, Sydney, Australia
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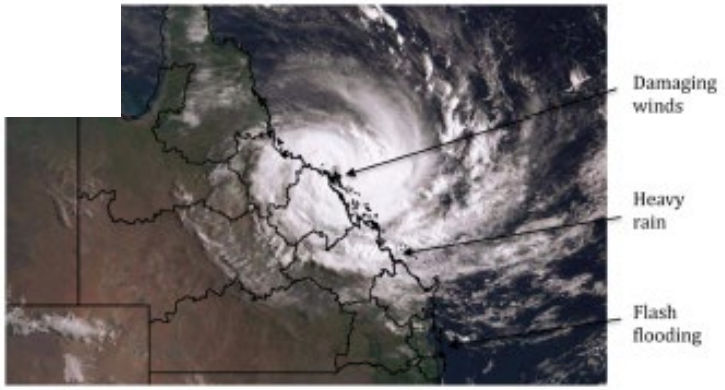
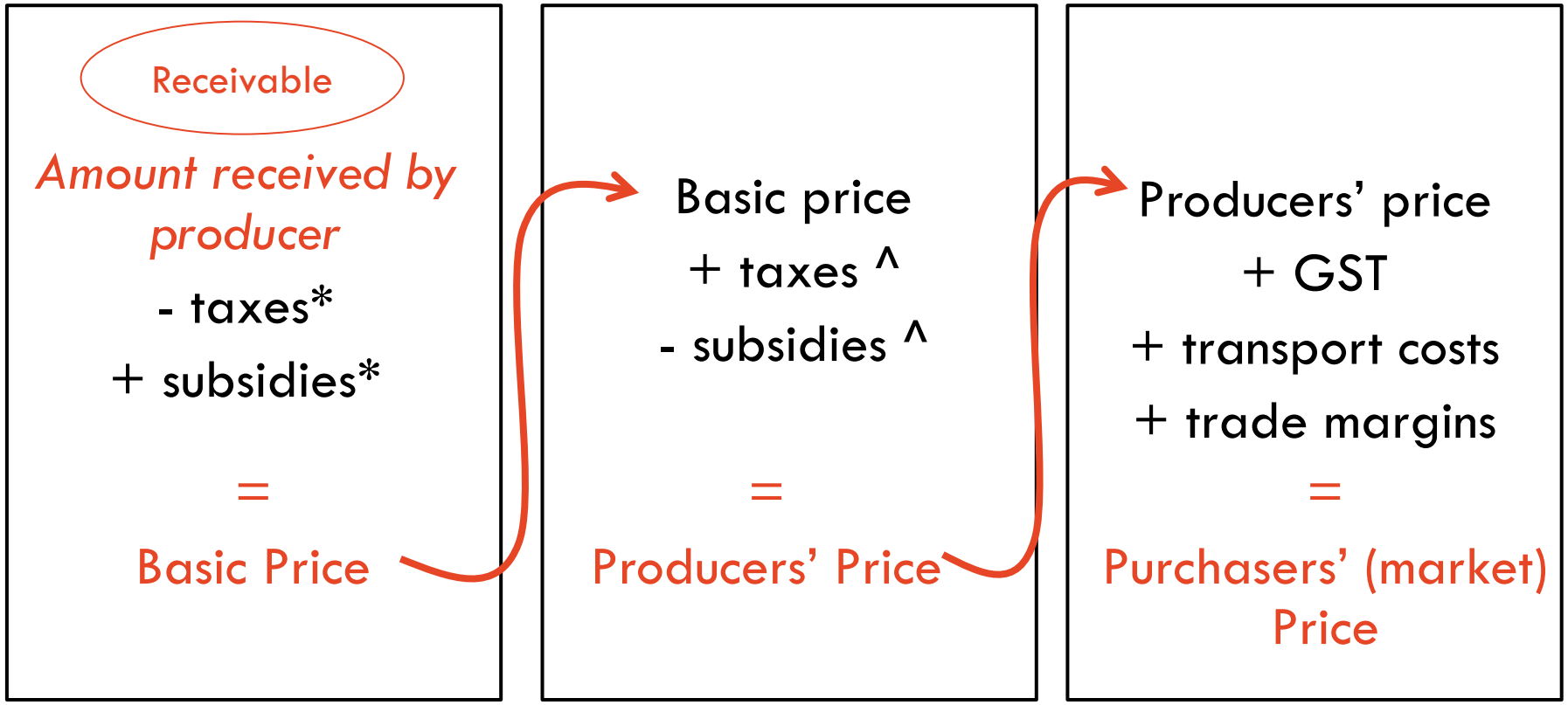


Figure 1. Geographical distribution of value-added loss caused by Tropical Cyclone Debbie. Value-added (VA) loss is expressed as $\Delta VA = q_{VA}(\mathbf{I}-\mathbf{A})^{-1}\Delta\mathbf{y}$. A comparison of our results (top, $-\log_{10}(\Delta VA)$, with ΔVA in AUD m) with a satellite image of the cyclone (ABC, 2017) (bottom) shows losses in northern Queensland regions as a direct consequence of the destructive winds and losses in southern Queensland and northern NSW as a result of heavy rain and floods occurring in the cyclone's wake. Region acronyms: RemNSW is the

Lenzen, M., et al. (2019). "Economic damage and spillovers from a tropical cyclone." Natural hazards and earth system sciences **19**(1): 137-151.

Valuation and different pricing options in the SNA



* taxes and subsidies on the production process (eg payroll tax, land tax, jobkeeper)

^ taxes and subsidies on the product itself (eg tobacco & alcohol excises)

Where are those prices used in an IO table?

	AUD$	Primary		Secondary		Tertiary			Final Demand				
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	TOTAL OUTPUT
Primary	Agriculture	2990	856	2878	2277	2968	510	12479					20538
	Mining	350				7	56	6134					15152
Secondary	Construction	1549				8	2278	11936					24384
	Manufacturing	956				5	621	15443					27614
Tertiary	Retail	822				5	779	4485					10162
	Banking	2077	716	568	5156	2236	354	11107					13599
	Subtotal T	8744	8566	14603	16554	8519	4598						
Value-added	Household	5789				9	1386						
	Government	3048				0	5865						
	Imports	2957				4	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

Valued at Basic price (i.e. excl. all taxes and subsidies)

Includes taxes and subsidies

Valued at Producers' price for all sectors excl. 'margin' sectors

* GST is accounted for in the Y matrix only for Households, since they are not able to claim a deduction for the GST paid, unlike businesses. This means that the Producers' price in the Y matrix is a 'hybrid' since it sometimes does include taxes on products (See SNA clause 6.63).

How are our purchases accounted for?

The monetary values included in an input-output table are derived using multiple sources of data to identify the total value of:

- prices paid by final consumers
- taxes and subsidies
- the components of price that are directed towards margin sectors

Margin sectors are those that facilitate the transfer of products from the manufacturer to the purchaser, without changing the products. These include the retail, wholesale, and transport sectors and sub-sectors.

The following slides are illustrative in nature to help understand how this price deconstruction occurs. They start from an individual product purchase, which is not how this works in practice for input-output table development. The base from which calculations are made is also illustrative and has been simplified whenever possible.

How are our purchases accounted for?



Assumes:
1.25% transport cost #
20% trade margin #
5% production taxes *
2% food industry subsidy *
No GST
No product tax
No product subsidy

Receivable	$3.25
- Taxes*	$0.16
+ subsidies*	$0.06
Basic price	$3.15
+ Taxes^	$0.00
- subsidies^	$0.00
Producer price	$3.15
+ GST#	$0.00
+ Transport costs#	$0.05
+ Trade margin#	$0.80
Market price	$4.00

Govt revenue =$0.10

calculated as % of market price
* taxes and subsidies on the production process (calculated as % of Basic price)
^ taxes and subsidies on the product itself (calculated as % of Market Price)

How are our purchases accounted for?



Assumes:

1.25% transport cost #

20% trade margin #

5% production taxes *

10% GST #

No product tax

No product subsidy

Receivable	$12.26
- Taxes*	$0.58
+ subsidies*	$0.00
Basic price	$11.68
+ Taxes^	$0.00
- subsidies^	$0.00
Producer price	$11.68
+ GST#	$1.70
+ Transport costs#	$0.21
+ Trade margin#	$3.40
Market price	$16.99

Govt revenue =$2.28

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

How are our purchases accounted for?



Receivable	$15.49
- Taxes*	$0.82
+ subsidies*	$1.63
Basic price	$16.30
+ Taxes^	$53.7
- subsidies^	$0.00
Producer price	$70.00
+ GST#	$10.00
+ Transport costs#	$0.00
+ Trade margin#	$20.00
Market price	$100.00

Assumes:
20% trade margin #
53.7% fuel excise ^@
5% production taxes *
10% fuel security subsidy *
10% GST #
No product subsidy
No transport cost to consumer

Govt revenue =$62.89

calculated as % of market price
* taxes and subsidies on the production process (calculated as % of Basic price)
^ taxes and subsidies on the product itself (calculated as % of Market Price)
@ for the purposes of this example, we will apply the excise to price rather than volume

How do our purchases show up?

Note: we swap the 'direction' of operators when swapping direction of calculation

Receivable	$

- Taxes*	$
+ subsidies*	$
Basic price	$

?

+ Taxes^	$
- subsidies^	$
Producer price	$

+ GST#	$1.00
+ Transport costs#	$0.00
+ Trade margin#	$2.00
Market price	$10.00

start from here

Assumes:

20% trade margin #

5% production taxes *

10% GST #

No transport cost to consumer

No product tax

No product subsidy

No subsidies on production

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

How do our purchases show up?

Note: we swap the 'direction' of operators when swapping direction of calculation

Receivable	$
- Taxes*	$
+ subsidies*	$
Basic price	$
+ Taxes^	$0.00
- subsidies^	$0.00
Producer price	$7.00
+ GST#	$1.00
+ Transport costs#	$0.00
+ Trade margin#	$2.00
Market price	$10.00

?

Assumes:

20% trade margin #

5% production taxes *

10% GST #

No transport cost to consumer

No product tax

No product subsidy

No subsidies on production

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

How do our purchases show up?

Note: we swap the 'direction' of operators when swapping direction of calculation

?

Receivable	$
- Taxes*	$0.35
+ subsidies*	$0.00
Basic price	$7.00
+ Taxes^	$0.00
- subsidies^	$0.00
Producer price	$7.00
+ GST#	$1.00
+ Transport costs#	$0.00
+ Trade margin#	$2.00
Market price	$10.00



Assumes:

20% trade margin #

5% production taxes *

10% GST #

No transport cost to consumer

No product tax

No product subsidy

No subsidies on production

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

How do our purchases show up?

?

Receivable	$7.35
- Taxes*	$0.35
+ subsidies*	$0.00
Basic price	$7.00
+ Taxes^	$0.00
- subsidies^	$0.00
Producer price	$7.00
+ GST#	$1.00
+ Transport costs#	$0.00
+ Trade margin#	$2.00
Market price	$10.00

Assumes:

20% trade margin #

5% production taxes *

10% GST #

No transport cost to consumer

No product tax

No product subsidy

No subsidies on production

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

How do our purchases show up?

		Check
Receivable	$7.35	7.35
- Taxes*	$0.35	- 0.35
+ subsidies*	$0.00	
Basic price	$7.00	= 7.00
+ Taxes^	$0.00	- 0.00
- subsidies^	$0.00	
Producer price	$7.00	= 7.00
+ GST#	$1.00	
+ Transport costs#	$0.00	+ 3.00
+ Trade margin#	$2.00	
Market price	$10.00	= 10.00

?

Assumes:
20% trade margin #
5% production taxes *
10% GST #
No transport cost to consumer
No product tax
No product subsidy
No subsidies on production

calculated as % of market price
* taxes and subsidies on the production process (calculated as % of Basic price)
^ taxes and subsidies on the product itself (calculated as % of Market Price)

How do our purchases show up?

Govt revenue =$1.35

Receivable	$7.35

?

- Taxes*	$0.35
+ subsidies*	$0.00
Basic price	$7.00

+ Taxes^	$0.00
- subsidies^	$0.00
Producer price	$7.00

+ GST#	$1.00
+ Transport costs#	$0.00
+ Trade margin#	$2.00
Market price	$10.00

Assumes:
20% trade margin #
5% production taxes *
10% GST #
No transport cost to consumer
No product tax
No product subsidy
No subsidies on production

calculated as % of market price
* taxes and subsidies on the production process (calculated as % of Basic price)
^ taxes and subsidies on the product itself (calculated as % of Market Price)

Input-output tables and valuation

	AUD$	Primary		Secondary		Tertiary			Final Demand				
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	TOTAL OUTPUT
Primary	Agriculture	2990	856	2878	2277	2968	510	12479	2	2344	5713	8059	20538
	Mining	350	1647	1478	2496	107	56	6134	1	2611	6406	9018	15152
Secondary	Construction	1549	1029	3054	2568	1458	2278	11936	3789	4785	3874	12448	24384
	Manufacturing	956	3956	5064	3521	1325	621	15443	4756	3256	4159	12171	27614
Tertiary	Retail	822	362	1561	536	425	779	4485	2564	1857	1256	5677	10162
	Banking	2077	716	568	5156	2236	354	11107	957	654	881	2492	13599
	Subtotal T	8744	8566	14603	16554	8519	4598						
Value-added	Household	5789	2419	5198	7854	1189	1386						
	Government	3048	3802	2127	2591	120	5865						
	Imports	2957	365	2456	615	334	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

T

Y

V

Input-output table valuation – breaking it down (T)

Basic prices

	AUD$	Primary		Secondary		Tertiary			Final Demand				TOTAL OUTPUT
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	
Primary	Agriculture	2990	856	2878	2277	2968	510	12479	2	2344	5713	8059	20538
	Mining	350	1647	1478	2496	107	56	6134	1	2611	6406	9018	15152
Secondary	Construction	1549	1029	3054	2568	1458	2278	11936	3789	4785	3874	12448	24384
	Manufacturing	956	3956	5064	3521	1325	621	15443	4756	3256	4159	12171	27614
Tertiary	Retail	822	362	1561	536	425	779	4485	2564	1857	1256	5677	10162
	Banking	2077	716	568	5156	2236	354	11107	957	654	881	2492	13599
	Subtotal T	8744	8566	14603	16554	8519	4598						
Value-added	Household	5789	2419	5198	7854	1189	1386						
	Government	3048	3802	2127	2591	120	5865						
	Imports	2957	365	2456	615	334	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

T

Y

V

Transactions for the **T** matrix are accounted for (or valued) in basic prices since taxes and subsidies on products may not be evenly applied across all sectors (eg Agriculture & Mining are exempt from fuel excise)

Input-output table valuation – breaking it down (Y)

Purchasers' prices

	AUD$	Primary		Secondary		Tertiary			Final Demand				TOTAL OUTPUT
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	
Primary	Agriculture	2990	856	2878	2277	2968	510	12479	2	2344	5713	8059	20538
	Mining	350	1647	1478	2496	107	56	6134	1	2611	6406	9018	15152
Secondary	Construction	1549	1029	3054	2568	1458	2278	11936	3789	4785	3874	12448	24384
	Manufacturing	956	3956	5064	3521	1325	621	15443	4756	3256	4159	12171	27614
Tertiary	Retail	822	362	1561	536	425	779	4485	2564	1857	1256	5677	10162
	Banking	2077	716	568	5156	2236	354	11107	957	654	881	2492	13599
	Subtotal T	8744	8566	14603	16554	8519	4598						
Value-added	Household	5789	2419	5198	7854	1189	1386						
	Government	3048	3802	2127	2591	120	5865						
	Imports	2957	365	2456	615	334	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

T

Y

V

The grand total of the **Y** matrix is at purchasers' prices, but at a sector level demand is accounted for (or valued) at producers' prices*. The transport and retail/wholesale trade sector data are derived from the difference between purchasers' and producers' prices in the other sectors

* GST is accounted for in the Y matrix only for Households, since they are not able to claim a deduction for the GST paid, unlike businesses. This means that the Producers' price in the Y matrix is a 'hybrid' since it sometimes does include taxes on products (See SNA clause 6.63).

Input-output table valuation – breaking it down (v)

	AUD$	Primary		Secondary		Tertiary			Final Demand				
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	TOTAL OUTPUT
Primary	Agriculture	2990	856	2878	2277	2968	510	12479	2	2344	5713	8059	20538
	Mining	350	1647	1478	2496	107	56	6134	1	2611	6406	9018	15152
Secondary	Construction	1549	1029	3054	2568	1458	2278	11936	3789	4785	3874	12448	24384
	Manufacturing	956	3956	5064	3521	1325	621	15443	4756	3256	4159	12171	27614
Tertiary	Retail	822	362	1561	536	425	779	4485	2564	1857	1256	5677	10162
	Banking	2077	716	568	5156	2236	354	11107	957	654	881	2492	13599
	Subtotal T	8744	8566	14603	16554	8519	4598						
Value-added	Household	5789	2419	5198	7854	1189	1386						
	Government	3048	3802	2127	2591	120	5865						
	Imports	2957	365	2456	615	334	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

T

Y

V

The **v** matrix includes taxes (and subsidies). These are a required input to production – the sector 'can't' produce its output without them.

Input-output table valuation – breaking it down (input)

AUD$		Primary		Secondary		Tertiary			Final Demand				TOTAL OUTPUT
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	
Primary	Agriculture	2990	856	2878	2277	2968	510	12479	2	2344	5713	8059	20538
	Mining	350	1647	1478	2496	107	56	6134	1	2611	6406	9018	15152
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	Government	3048	3802	2127	2591	120	5865						
	Imports	2957	365	2456	615	334	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

T

Y

V

Input includes taxes (and subsidies) as a line item in the **v** block

Input-output table valuation – breaking it down (output)

AUD$		Primary		Secondary		Tertiary			Final Demand				TOTAL OUTPUT
		Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Subtotal T	Households	Government	Exports	Subtotal Y	
Primary	Agriculture	2990	856	2878	2277	2968	510	12479	2	2344	5713	8059	20538
	Mining	350	1647	1478	2496	107	56	6134	1	2611	6406	9018	15152
Secondary	Construction	1549	1029	3054	2568	1458	2278	11936	3789	4785	3874	12448	24384
	Manufacturing	956	3956	5064	3521	1325	621	15443	4756	3256	4159	12171	27614
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	Banking												13599
	Subtotal T	8744	8566	14603	16554	8519	4598						
Value-added	Household	5789	2419	5198	7854	1189	1386						
	Government	3048	3802	2127	2591	120	5865						
	Imports	2957	365	2456	615	334	1750						
	Subtotal V	11794	6586	9781	11060	1643	9001						
	TOTAL INPUT	20538	15152	24384	27614	10162	13599						

T

Y

V

Output includes taxes (and subsidies) in producers' prices

Mangoes

Workings



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Receivable	$
- Taxes*	$
+ subsidies*	$
Basic price	$
+ Taxes^	$
- subsidies^	$
Producer price	$
+ GST#	$0.50
+ Transport costs#	$0.00
+ Trade margin#	$0.75
Market price	$5.00

$5 x 10%

NA

$5 x 15%

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Mangoes

Workings



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Receivable	$
- Taxes*	$
+ subsidies*	$
Basic price	$
+ Taxes^	$0.00
- subsidies^	$0.15
Producer price	$3.75
+ GST#	$0.50
+ Transport costs#	$0.00
+ Trade margin#	$0.75
Market price	$5.00

NA

$5 x 3%

$5-$0.75-$0.5

$5 x 10%

NA

$5 x 15%

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Mangoes

Workings



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Receivable	$	
- Taxes*	$0.04	*$3.90 x 1%*
+ subsidies*	$0.00	*NA*
Basic price	$3.90	*$3.75 + $0.15*
+ Taxes^	$0.00	*NA*
- subsidies^	$0.15	*$5 x 3%*
Producer price	$3.75	*$5-$0.75-$0.5*
+ GST#	$0.50	*$5 x 10%*
+ Transport costs#	$0.00	*NA*
+ Trade margin#	$0.75	*$5 x 15%*
Market price	$5.00	

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Mangoes



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Workings

Receivable	$3.94	$3.90 + $0.04
- Taxes*	$0.04	$3.90 x 1%
+ subsidies*	$0.00	NA
Basic price	$3.90	$3.75 + $0.15
+ Taxes^	$0.00	NA
- subsidies^	$0.15	$5 x 3%
Producer price	$3.75	$5-$0.75-$0.5
+ GST#	$0.50	$5 x 10%
+ Transport costs#	$0.00	NA
+ Trade margin#	$0.75	$5 x 15%
Market price	$5.00	

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Mangoes



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Producer receivable

Receivable	$3.94
- Taxes*	$0.04
+ subsidies*	$0.00
Basic price	$3.90
+ Taxes^	$0.00
- subsidies^	$0.15
Producer price	$3.75
+ GST#	$0.50
+ Transport costs#	$0.00
+ Trade margin#	$0.75
Market price	$5.00

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Mangoes

Govt revenue =$0.39



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Receivable	$3.94
- Taxes*	$0.04
+ subsidies*	$0.00
Basic price	$3.90
+ Taxes^	$0.00
- subsidies^	$0.15
Producer price	$3.75
+ GST#	$0.50
+ Transport costs#	$0.00
+ Trade margin#	$0.75
Market price	$5.00

Govt revenue

= 0.04 - 0.15 + 0.50

0.39

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Mangoes



Choose a product from the Agriculture sector

Assumes:

15% trade margin #

1% production taxes *

3% fruit industry subsidy ^

10% GST #

No transport cost to consumer

No product tax

No subsidies on production

Receivable	$3.94
- Taxes*	$0.04
+ subsidies*	$0.00
Basic price	$3.90
+ Taxes^	$0.00
- subsidies^	$0.15
Producer price	$3.75
+ GST#	$0.50
+ Transport costs#	$0.00
+ Trade margin#	$0.75
Market price	$5.00

Retailer margin income

calculated as % of market price

* taxes and subsidies on the production process (calculated as % of Basic price)

^ taxes and subsidies on the product itself (calculated as % of Market Price)

Tutorial



THE UNIVERSITY OF
SYDNEY

Tutorial exercise part I – the role of a local council

In small groups discuss:

1. Which local council area do you live in?
   - If you aren't based in Australia, choose an Australian council that looks interesting to you, or you could choose the City of Sydney Council which is the one the University of Sydney is part of.

2. What services is your local council responsible for delivering?

3. What are some of the impacts of climate change that might be felt by your local council?

4. Is your local council doing anything to mitigate or adapt to climate change?

Useful links: https://www.olg.nsw.gov.au/public/find-your-local-council
https://peo.gov.au/understand-our-parliament/how-parliament-works/three-levels-of-government/three-levels-of-government-governing-australia/

Tutorial exercise part II

Using the IO table created in Week 1, choose 3 of the sectors in your table. For each sector, choose a product (real or imaginary).

Carry out calculations for each product showing how the market price is broken down into:

1. Producer price
2. Basic price
3. Receivable

You can make any assumptions you want on the values for each line (eg tax rate, margin, subsidies), just make sure you note what those assumptions are and that your workings are visible.

Homework for Week 3

Homework week 3 (Due: September 9, 2026)

Decomposing prices in IO tables

Using your 14-sector IO table, choose 1 sector in your table. Choose a product that is an output of that sector and allocate that product a market price (this can be the actual market price of that product, or you can make it up).

Carry out calculations for that product showing how the **market price (or purchasers' price)** is broken down into:

1. Producers' price
2. Basic price
3. Receivable

You can make any assumptions you want on the value of the components used for your calculations (e.g. tax rates, margins, subsidies), just make sure that you note what those assumptions are. Make sure that the workings for your calculations are available, ideally by using equations in Excel.

EXPLANATION: Explain the concept of valuation in input-output tables (150 words). Comment on why different countries have their own input-output tables (about 150 words).

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- Input-output tables form the basis of many economic indicators
- Statistical agencies collate the data from many sources
- Basic Price -> Producers' Price -> Purchasers' Price
- Difference between taxes on production and taxes on products

More to come:

- The A and L matrices
- Footprint assessments